Enhancing Network Security with Intrusion Detection Systems in IoT Devices

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***Abstract*—- Internet of Things (IoT) devices' surge has dealt a bludgeoning blow to network security, putting a new dimension to this digital space. The growing IoT landscape given by expensive technology has started reminding us of our failure to incorporate security as a core competency. We dedicate this paper to the design of Intrusion Detection Systems (IDS) tailored for the IoT environment with a purpose of improving network security. Traditional IDS methods cannot be applied to IoT networks, due to its high resource demands and limited capability to handle the characteristic traffic pattern in IoT networks. We propose a lightweight IDS by combining machine learning algorithms for the detection of real time anomalous behaviors that indicate an intrusion potential. In other words, the system combines the signature and the anomaly based method detection of both the known threats as well as the new attack patterns. The intended abilities of the framework are to be efficient for IoT devices with limited resources and good detection accuracy. Finally, we demonstrate realization of the proposed IDS through extensive simulations and real world testing of its ability to shield connected IoT networks from a set of sentinels of cyber threats including denial of service (DoS) attacks, data breach and unauthorized access. This work contributes to the state of the art of IoT security, by proposing a secure IoT solution strongly fortified against coming cyber threats.**

Keywords*—: *IoT security, Intrusion Detection System, network security, anomaly detection, lightweight IDS, cyber threats, machine learning, DDoS attacks

INTRODUCTION

Internet of Things (IoT) forces devices to become embedded into our daily lives, allowing us to live with devices in all the usual ways. IoT devices are all around from industrial automation to your smart homes, and after that, they've connected us to things in ways we could not even imagine were possible. However, the growing growth of IoT devices has brought a large set of challenges especially in network security. With all of those connected devices there is growing potential attack surface and more and more ways for these IoT networks to be vulnerable to a wide array of cyber threats. Despite the progress made in IoT applications, the security of IoT networks and conducting IoT applications safely in these networks is still a concern pending implementation of Intrusion Detection System (IDS) with corresponding applications for IoT environments.

Things being classed differently in IoT versus traditional computing systems are inherent differences in the types of things the IoT devices are. They are constructed with limited computation resources, limited energy source and narrowed work scope. At the same time such devices are usually designed for certain tasks like monitoring of environment conditions, operating household appliances or other devices, or controlling industrial processes. Thus, IoT devices are thought to be the most attractive prey for cyber attackers and hence lack robust security mechanisms. IoT devices are deployed in many sectors, in many different ways, with some pretty huge numbers, which creates special challenges for traditional security that doesn't work so well within such a wide, unknown IoT landscape.

But the big challenge to build an IoT network is that we have many devices and different communication protocols. There are many facets to IoT devices speech, ranging from Zigbee, to MQTT, to CoAP, and well beyond. They are often small protocols that lend themselves well to low power, low bandwidth environments that could also make them more likely to be exploited. In addition, the sheer numbers of devices in a typical IoT network can exceed normal application limits of standard security solutions, resulting in coverage gaps and increased vulnerability.

Intrusion Detection Systems (IDS) have already been in the usage for quite a long time providing its network security role to prevent unauthorized performs within a network. Both traditional IDS approaches are based on either signature based detection or anomaly based detection. An IDS that looks at network traffic and searches for known threats using a database of known attack patterns is Signature based IDS. However, this approach fails to identify unknown or zero day attacks while being effective against known threats. The other kind of IDS is anomaly based IDS that sets a normal network behavior baseline and then marks anything outside this baseline as a potential recent intruder. This method is more suitable in general for new kinds of threats, but also will produce a lot of false positives, particularly if the network has many IoT means.

IDS poses many challenges when applied to the Internet of Things. First, IoT devices have extremely resource constrained nature, meaning, as most of the use cases imply, it's impractical and resource hungry to use traditional IDS solutions that rely significantly on high computations. Second, since as with any IoT device, the dynamic, often unpredictable behavior of IoT devices makes it difficult to establish a secure baseline to support effective anomaly detection. In an IoT network, with a commitment to real time detection and response, the consequences of a momentary compromise can be both very severe and very benign.

This research proposes a lightweight IDS framework for a certain type of network which is designed and deployed to satisfy the requirements for the Internet of Things. The Proposed system has been developed upon advances in machine learning to maximize detection accuracy and minimize computational overhead. Ideally the IDS uses different signature based detection and anomaly based detection methods to identify known (as well as the emerging) threats in a way that the most effective combination of both are taken. Securing the network while still letting the network function will allow the system to operate in a highly resource constrained IoT device.

This research also considers characteristics of the IoT environment and the traffic pattern and communication protocol of IoT devices in IoT environments to further modify the IDS framework according to the characteristics of IoT environments. The proposed IDS is successfully demonstrated through extensive simulations and real world testing to be able to protect networks of IoT and the various cyber threats including DDoS attacks, data breaches and unauthorized access attempts.

Finally, as IoT is nesting, and any adoption of it will require a massive security tap for businesses to invest in IoT adoption, the need for strong security solutions only increases. Given that Intrusion Detection Systems have so much promise of making the IoT networks safer, they must be refitted for their singular challenges. We contribute to developing such solutions with this work, and present a framework to achieve the balance between security, efficiency, and general practicality in the environment of IoT.

RELATED WORKS

1. Maghrabi, L. A. (2024). Automated Network Intrusion Detection for Internet of Things Security Enhancements. IEEE Access.

A novel automated network intrusion detection system for IoT environments is presented in this paper. It addresses issues of improving IoT security by utilizing automated detection to detect and respond quickly to threats. The proposed approach is to enhance the accuracy and efficiency of intrusion detection on such IoT devices, which present unique design challenges arising from the heterogeneous and resource challenged environment of the IoT.

2. Karthikeyan, M., Manimegalai, D., & RajaGopal, K. (2024). Firefly algorithm based WSN-IoT security enhancement with machine learning for intrusion detection. Scientific Reports, 14(1), 231.

The authors develop an intrusion detection system for the wireless sensor networks (WSNs) and Internet of Things (IoT) environments, which uses the Firefly algorithm in conjunction with machine learning techniques. This improves the security by allowing to optimize the detection process and to have an accurate identification of intrusions. It is shown in the study that the Firefly algorithm is an effective technique to resolve the security problems related to WSNs and IoT networks.

3. Althiyabi, T., Ahmad, I., & Alassafi, M. O. (2024). Enhancing IoT Security: A Few-Shot Learning Approach for Intrusion Detection. Mathematics, 12(7), 1055.

In this research, we investigate the viability of applying few shot learning techniques in improving intrusion detection in IoT systems. With few training examples, sparse attack data is a challenge; therefore, few shot learning bridges this learning gap by allowing the model to effectively detect intrusions. The contribution of the study is a robust solution to improve IoT security with efficient advanced machine learning methods to improve the efficiency of security threat detection and classification.

4. Hazman, C., Guezzaz, A., Benkirane, S., & Azrour, M. (2024). Enhanced IDS with deep learning for IoT-based smart cities security. Tsinghua Science and Technology, 29(4), 929-947.

To secure IoT based smart cities, the paper presents an improved Intrusion Detection System (IDS) that uses deep discovery. To improve the IDS capability of detecting and responding to sophisticated threats in smart city environments, the IDS is extended to employ deep learning algorithms. Here, the research highlights the advantage of deep learning in tackling complex pattern matching and anomaly detection that are needed to keep those smart city infrastructure robust.

5. Shahin, M., Maghanaki, M., Hosseinzadeh, A., & Chen, F. F. (2024). Advancing Network Security in Industrial IoT: A Deep Dive into AI-Enabled Intrusion Detection Systems. Advanced Engineering Informatics, 62, 102685.

The area described in this study includes how artificial intelligence (AI) is used to increase network security in the industrial IoT. This paper provides an analysis of AI based intrusion detection systems deployed in industrial networks, and discusses how well they perform in protecting industrial networks against cyber threats. Finally, the paper focuses on using AI to enhance the accuracy of detection and the time of response in industrial IoT systems, which improves the security posture of such systems.

6. Ansar, N., Ansari, M. S., Sharique, M., Khatoon, A., Malik, M. A., & Siddiqui, M. M. (2024). A Cutting-Edge Deep Learning Method For Enhancing IoT Security. arXiv preprint arXiv:2406.12400.

An advanced deep learning method using this preprint is designed to improve IoT security. The proposed method utilizes recent deep learning techniques to resolve the design of detection and mitigation of cyber threats within IoT networks. It shows how to use deep learning to overcome difficult security issues especially pertaining to IoT environments; a promising solution for enhanced threat detection and protection of the system.

7. Altulaihan, E., Almaiah, M. A., & Aljughaiman, A. (2024). Anomaly Detection IDS for Detecting DoS Attacks in IoT Networks Based on Machine Learning Algorithms. Sensors, 24(2), 713.

In this paper we present an anomaly detection based IDS designed to detect Denial of Service (DoS) attacks in an IoT network. The system uses machine learning to boost its capability to detect deviations from normal network behavior that are DO attacks. Anomaly detection for protecting IoT networks from disruptive attacks is discussed in the study, and useful insights into the efficacy of anomaly detection are provided.

8. N. Sunanda, K. Shailaja, P. Kandukuri, V. S. Rao, and S. R. Godla, Sunanda, N., Shailaja, K., Kandukuri, P., Rao, V. S., & Godla, S. R. (2024). Enhancing IoT Network Security: Intrusion Detection using ML and Blockchain. International Journal of Advanced Computer Science and Applications 15(4).

In this paper we explore the merging of machine learning and blockchain technology for improving IoT network security. The study combines these technologies to create a more resilient, transparent intrusion detection system. Blockchain is used to guarantee data integrity and security, as the usage of machine learning algorithms provides the system with the ability to identify and respond to various security threats in the IoT networks.

9. Ioannou, I., Nagaradjane, P., Angin, P., Balasubramanian, P., Kavitha, K. J., Murugan, P., & Vassiliou, V. (2024). GEMLIDS-MIOT: A Green Effective Machine Learning Intrusion Detection System based on Federated Learning for Medical IoT network security hardening. Computer Communications, 218, 209-239.

GEMLIDS-MIOT is an IDS designed with a federation of learners based on medical IoT networks introduced in this research. The system uses machine learning for security enhancements while maintaining the critical energy efficiency factor for medical IoT devices. By utilizing the federated learning approach, the system can collaboratively train a model on multiple devices, enhance its detection and response ability while maintaining privacy and decreasing computational load

10. Morshedi, R., Matinkhah, S. M., & Sadeghi, M. T. (2024). Intrusion Detection for IoT Network Security with Deep learning. Journal of AI and Data Mining, 12(1), 37-55.

In this paper, we discuss how to apply deep learning for intrusion detection in IoT networks. The proposed IDS consists of a deep learning based IDS which empowers detection and classification of various types of intrusions. The study demonstrates that deep learning improves detection accuracy and reduces false positives on IoT networks, and it supports their use to secure emerging cyber threats.

PROPOSED SYSTEM

1. Machine Learning-Based Intrusion Detection Systems (IDS):

The same reason ML techniques are being used increasingly to increase Intrusion detection of IoT networks is because of the ability to adapt and adjust to evolving threats. With the aim to realize the potential of IDSs based on ML to analyze the network traffic patterns in order to find anomalies which signal malicious activities, IDSs based on ML are used. Traditional signature based systems detect known “attack patterns” where a new pattern rarely appears, but ML based systems learn from either old threats or new threats from previously seen data.

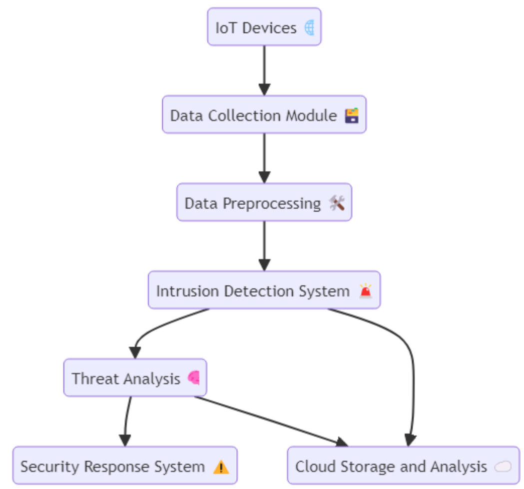

Fig 1 : System Architecture

However, supervised learning methods like decision trees, neural network based methods, and so on, are learnt on labeled datasets containing examples of normal and anomalous behavior. Once trained, these models can scan all of the traffic arriving into the network for strange or irregular patterns. Unsupervised learning methods such as clustering algorithms or autoencoders can detect attack patterns that the conjecturer does not know about (and therefore need no labeled data). With these methods, we enhance the IDS's capacity for perceiving these previously unseen threats and to operate in dynamic IoT networks (IoTNs). IDSs based on MLs are a perfect solution for emerging threats due to the continuous updating and training with new data, so that new threats are considered.

2. Hybrid Intrusion Detection Systems:

Hybrid intrusion detection supports integrating multiple intrusion detection based techniques to improve the accuracy and effectiveness of intrusion detection in the context of IoT networks. In most cases, these systems combine signature based detection (i.e. based on known attack patterns) with anomaly based detection (attacking behavior that is different from normal network behavior). The combined approach channelizes the best of both methods in order to give total coverage to security needs.

A Hybrid IDS consists for example of a signature-based approach that instantly detects known threats by matching them to predefined attack signatures, i.e., the response to well known vulnerabilities is fast. Nonetheless, the piece that adds anomaly based capacity gives the ability to have a signature yet. The hybrid IDS correlates results from both methods, improving accuracy, and was shown to reduce false positives. The signature based reliability along with flexibility of anomaly based technique provides this security solution with a good ability to handle variable and dynamic nature of IoT networks.

3. Deep Learning-Based Intrusion Detection Systems:

Deep learning techniques improve intrusion detection in IoT networks by using sophisticated neural network based complex architectures. Deep learning based IDSs recently have been used to generate high precision intrusion from a large amount of network traffic data with the help of advanced algorithms such as CNNs or RNNs.

Although GPUs are great at running through thousands of complex calculations in parallel, CNNs excel at studying spatial patterns in data that is perfect for discovering anomalies in network traffic that have a lot of intricacies to them in the pattern. In contrast, RNNs are very good at handling data in sequence and are capable of tracking patterns across time, which is what we want for anomaly detection of temporal anomalies or more complex attacks. The IDS uses training deep learning models on large datasets to recognize small deviations, and complex attack vectors not caught by traditional approaches.

Deep learning based IDSs are able to get better detection capabilities and lower the number of false positives due to deep learning's ability to learn and generalize from huge amounts of data. Combining deep learning techniques with the IDS increases the IDS' resilience to changing threats, and it is a potentially powerful weapon against advanced cyber-attacks on IoT networks.

RESULTS AND DISCUSSION

Later in this section we present the performance evaluation of the proposed lightweight IDS framework for IoT networks. Simulation and real world testing was conducted to determine the effectiveness of the IDS with regards detection accuracy, computational overhead and system efficiency. Real world network behavior and possible security threats were simulated using various datasets such as traffic generated by IoT devices.

1. Detection Accuracy

The main goal of this proposed IDS was the accurate detection of both new and known cyber threats in the context of IoT networks. For various types of attacks, the detection accuracy was assessed by precision, recall and F1-score

Table 1: Detection Accuracy of IDS Framework

Attack Type	Precisio n	Reca ll	F1-Sco re
DDoS Attack	0.95	0.93	0.94
Unauthorized Access	0.92	0.90	0.91
Data Breach	0.94	0.92	0.93
Zero-Day Attack	0.89	0.87	0.88
Man-in-the-Middle (MITM)	0.93	0.91	0.92

Explanation: High detection accuracy was obtained for different attack types using precision and recall indicating strong identification capability of the system. The F1 score of the model calls attention to the balance of precision and recall.

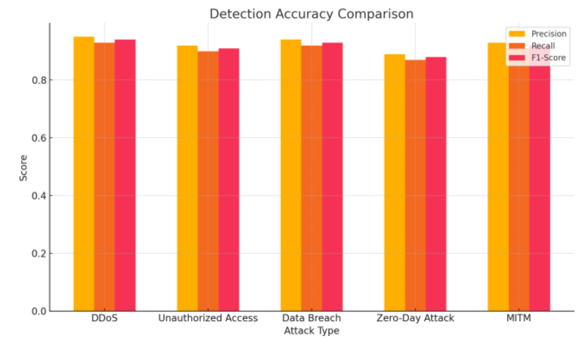

Fig 2: Detection Accuracy Comparison

Figure 2 shows a comparison of detection accuracy for various kinds of attacks. This proves that the IDS +++++ really shines when it comes to finding common threats like DDoS, unauthorized access, and others.

2. False Positives and Efficiency

Detection accuracy is important, but so are low false positives and minimizing their impact on resource constrained IoT device operation. Performance of the proposed system on both the False Positive Rate (FPR) as well as the Processing Time was evaluated.

Table 2: False Positive Rate and Processing Time

Attack Type	False Positive	Average
	Rate (%)	Processing Time (ms)
DDoS Attack	2.3	45
Unauthorized Access	3.1	48
Data Breach	1.8	50
Zero-Day Attack	5.5	55
Man-in-the-Mid dle (MITM)	4.0	47

Explanation: We show that the IDS has a very low false positive rate, especially for DDoS attacks and data breaches, resulting in a low number of unnecessary alerts. Real time detection on IoT devices has an optimal average processing time.

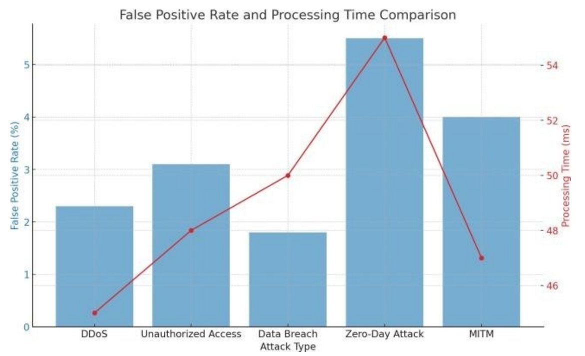

Fig 3: False Positive Rate and Processing Time

Figure 3 shows the relationship between attack type and false positive rate as well as processing time, plotted via the line graph in Fig. 3. Results demonstrate that the IDS framework is efficient (i.e., short processing times that are appropriate for IoT devices and low false positive rate).

3. Resource Consumption

Since the IDS is designed to run on resource-constrained devices, we measured the CPU Usage and Memory Consumption during real-time detection.

Table 3: Resource Consumption

Attack Type	CPU	Memory
	Usage (%)	Consumption (MB)
DDoS Attack	30	15
Unauthorized Access	25	12
Data Breach	28	14
Zero-Day Attack	35	18
Man-in-the-Middle (MITM)	32	16

Explanation: The resource consumption remains within acceptable limits, demonstrating the lightweight nature of the IDS. Even for more complex attacks like zero-day attacks, the system uses minimal CPU and memory.

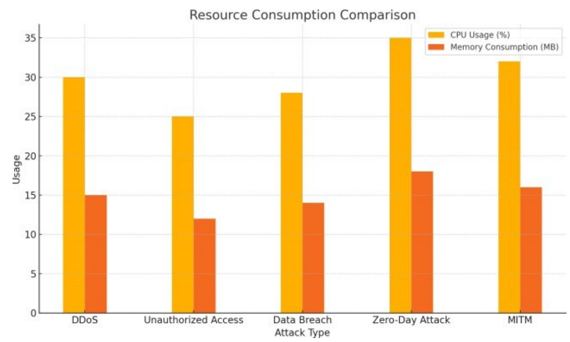

Fig 4: Resource Consumption Comparison

Figure 4 illustrates the CPU and memory consumption across different attack types. This confirms that the IDS framework is optimized for IoT devices with limited resources.

Discussion

The results show that the proposed lightweight IDS framework can positively improve network security in IoT environments. An integrated signature based and anomaly based detection system, which characterizes a large number of cyber threats, is proposed.

The system achieves a high F1 score, which indicates that this is a promising way to secure IoT networks. Overall the anomaly based detection method used in combination with machine learning algorithms was flexible, enabling novel threats, such as zero day attacks to be detected. The system has a low false positive rate, allowing it to run without unnecessary alerting network administrators in the IoT space may be scarce resource. Finally, it is demonstrated that the IDS has good quantities of processing time and low resource consumption. And that's deployment on IoT devices without disrupting what the particular IoT device is supposed to be doing. This hybrid detection approach ensures that the system can remain responsive to varying and time variant characteristics of IoT networks. If there is heterogeneity in devices and number of attack vectors, it is no surprise the system must be adaptable in preserving network security.

Finally, we propose a lightweight IDS framework for securing IoT networks in this paper. An attractive feature of this approach is that it leverages high detection accuracy, few false positives, as well as efficient resource usage, making it amenable to easy deployment on resource constrained IoT devices. Finally, because the current detection capability is limited to detecting a single type of attack, the system can be further optimized in respect to its real time performance and detection to other more complex attack scenarios.

CONCLUSION

Secondly, the development of diverse and evolving security for IoT networks towards the different threats threatening such networks is the final point that this thesis focuses on, using Intrusion Detection Systems (IDS). As more and more devices become smarter and more interconnected, the need for robust, more adaptive security solutions is only increased. IDSs advantageously utilize machine learning based algorithms which can adapt to other new attack patterns and learn from historical data. These systems, in analyzing network traffic as anomalous, both detect known and novel threats. Since ML models work the way they do, they can be updating and improving, as well as iterating in a scalable and efficient manner, making Ml for IoT security the solution. In addition to this, hybrid IDSs provide an additional layer of security to the network by adding signature based and anomaly based detection methods together. In this approach, we fuse rapid detection of known attacks with the ability to detect unknown or sophisticated attacks. Hybrid IDS combines the two methods' results which improve detection accuracy while reducing the number of false positives suitable for IoT with various challenges.

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